

Exploratory Data Analysis (EDA)

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Exploratory Data Analysis (EDA) is the process of **summarizing, visualizing, and understanding a dataset** to uncover patterns, spot anomalies, test hypotheses, and check assumptions before applying statistical modeling or machine learning.

Population Vs. Sample

A **population** is the complete set of all possible observations that can be made about a particular phenomenon or group.

It includes **every individual, object, or measurement** that fits a specific criterion.

Examples:

- All students enrolled in a university.
- Every manufactured part from a factory in one year.
- Every rainfall reading in India over 10 years.

Characteristics of a Population:

- Usually **large or infinite** in size.
- True values describing it are called **parameters**.

Common Population Parameters:

- Mean: $\mu = \frac{\sum X}{N}$
- Variance: $\sigma^2 = \frac{\sum (X - \mu)^2}{N}$
- Standard deviation: $\sigma = \sqrt{\sigma^2}$

Sample

A *sample* is a subset of the population selected for analysis or observation.

It should be **representative** of the population.

•Purpose:

- To make **inferences** about the population.
- To study large populations efficiently and economically.

•Example:

- 200 randomly selected students from a university to study average study hours.

•Characteristics of a Sample:

- Described by **statistics**, which are estimates of population parameters.

Common Sample Statistics:

- Mean: $\bar{X} = \frac{\sum X_i}{n}$
- Variance: $s^2 = \frac{\sum (X_i - \bar{X})^2}{n-1}$
- Standard deviation: $s = \sqrt{s^2}$

Comparing Populations

- Compare groups using **summary statistics**: mean, median, variance
- **Visual comparisons**: Boxplots, bar charts, violin plots
- **Statistical tests**:
 - t-test & z-test → Compare two groups
 - ANOVA → Compare more than two groups

Moments and Generating Functions

Moments

Moments are numerical measures that describe the *shape* and *characteristics* of a probability distribution.

There are two kinds of moments:

1.Raw Moments – about the origin

2.Central Moments – about the mean

(i) Raw Moments

- Formula:

$$\mu'_r = E(X^r) = \sum x_i^r P(x_i)$$

- These describe the distribution in relation to the origin (0).

(ii) Central Moments

- Formula:

$$\mu_r = E[(X - \mu)^r]$$

- These describe the distribution in relation to the **mean**.

Interpretation of Moments:

Moment	Formula	Description
1st Moment	$\mu'_1 = E(X) = \mu$	Mean (average value)
2nd Moment	$\mu_2 = E[(X - \mu)^2]$	Variance (spread of data)
3rd Moment	$\mu_3 = E[(X - \mu)^3]$	Skewness (symmetry of distribution)
4th Moment	$\mu_4 = E[(X - \mu)^4]$	Kurtosis (peakedness or flatness)

(iii) Skewness and Kurtosis

- Skewness (γ_1):

$$\gamma_1 = \frac{\mu_3}{\mu_2^{3/2}}$$

- Measures **asymmetry**.
- $\gamma_1 > 0$: Positively skewed (tail on right).
- $\gamma_1 < 0$: Negatively skewed (tail on left).
- Kurtosis (γ_2):

$$\gamma_2 = \frac{\mu_4}{\mu_2^2}$$

- Measures **peakedness**.
- $\gamma_2 = 3$: Normal curve (mesokurtic)
- $\gamma_2 > 3$: Leptokurtic (more peaked)
- $\gamma_2 < 3$: Platykurtic (flatter)

1. High kurtosis → heavy tails
2. Low kurtosis → flat distribution

Generating Functions

- **Probability generating functions (PGF)**: Summarize discrete distributions
- **Moment generating functions (MGF)**: Useful for deriving moments
- Often used for **theoretical understanding** of distributions

Measures of central tendency and variability

A. Measures of Central Tendency

They locate the "center" of a data distribution.

Measure	Formula / Definition	Key Property
Mean (Arithmetic Average)	$\bar{X} = \frac{\sum X_i}{n}$	Sensitive to all values (affected by outliers)
Median	Middle value when data are ordered	Not affected by extreme values
Mode	Most frequently occurring value	Best for categorical data
Geometric Mean	$G = (\prod X_i)^{1/n}$	For growth rates
Harmonic Mean	$H = \frac{n}{\sum(1/X_i)}$	For average of ratios or speeds

Example:

For data: 10, 12, 13, 15, 20

- Mean = 14
- Median = 13
- Mode = None (no repetition)

B. Measures of Variability (Dispersion)

They indicate how **spread out** or **scattered** the data are around the mean.

Measure	Formula	Interpretation
Range	$R = X_{max} - X_{min}$	Simplest measure of spread
Mean Deviation (MD)	$(MD = \frac{\sum X_i - \bar{X} }{n})$	$X_i - \bar{X}$
Variance	$s^2 = \frac{\sum (X_i - \bar{X})^2}{n-1}$	Measures overall dispersion
Standard Deviation (SD)	$s = \sqrt{s^2}$	Same unit as data
Coefficient of Variation (CV)	$\frac{s}{\bar{X}} \times 100$	Relative measure of dispersion

Example:

Data = [10, 12, 14, 16, 18]

- Mean = 14
- Variance = 8
- SD = $\sqrt{8} = 2.83$
- CV = $(2.83 / 14) \times 100 = 20.2\%$

Interpretation of Dispersion

- **Low dispersion:** Data are close to the mean → consistent performance.
- **High dispersion:** Data are spread out → variability or inconsistency.
- **Coefficient of Variation (CV) Comparison:**

When comparing two datasets of different units or magnitudes, the dataset with a **lower CV** is **more consistent**.

Statistical Foundations

Hypothesis Formulation and Testing

What is a Hypothesis?

A **hypothesis** is a statement or assumption about a population parameter that can be tested using statistical methods.

- It helps to make inferences from **sample data** about the **population**.

Types of Hypotheses

Type	Description	Example
Null Hypothesis (H_0)	Assumes no effect, no difference, or status quo .	H_0 : The average marks of male and female students are equal.
Alternative Hypothesis (H_1 or H_a)	Represents a new claim, effect, or difference .	H_1 : The average marks of male and female students are not equal.

Steps in Hypothesis Testing

1. State the Hypotheses

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

2. Select the Significance Level (α)

Common choices: 0.05, 0.01, or 0.10

It represents the *risk of rejecting a true null hypothesis*.

3. Select the Appropriate Test Statistic

Depends on data type, sample size, and known parameters.

Examples: *z-test*, *t-test*, *chi-square test*, *ANOVA*.

4. Compute the Test Statistic

$$z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

5. Find the Critical Region or p-value

Compare calculated value with table value (critical value) or directly use **p-value**.

6. Make a Decision

1. If $p\text{-value} < \alpha \rightarrow$ **Reject H_0** (significant difference).
2. If $p\text{-value} \geq \alpha \rightarrow$ **Fail to reject H_0** (no significant difference).

7. Draw a Conclusion

Interpret the result in practical terms (e.g., "there is a significant difference in...").

◆ What Are Z-Test and T-Test?

Both are **hypothesis tests** used to compare sample data with a population (or between two samples).

The difference mainly depends on **what you know about the population and sample size**.

◆ Z-Test

► When to use:

Use a **Z-test** when:

1. **Population standard deviation (σ) is known**, and
 2. **Sample size (n) ≥ 30** (large sample).
 3. Data follows a **normal distribution** (or approximately normal).
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► Formula:

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

► **Example:**

A company claims that the mean life of its batteries is **400 hours**, with $\sigma = 50$ hours.

A sample of **100 batteries** has a mean life of **390 hours**.

Test at $\alpha = 0.05$ whether the claim is true.

$$H_0 : \mu = 400 \quad \text{and} \quad H_1 : \mu \neq 400$$

$$Z = \frac{390 - 400}{50/\sqrt{100}} = \frac{-10}{5} = -2$$

For a **two-tailed test** at $\alpha = 0.05$,

critical $z = \pm 1.96$

Since $|-2| > 1.96$,

 **Reject H_0** — the average battery life is significantly different from 400 hours.

◆ T-Test

► When to use:

Use a **T-test** when:

1. **Population standard deviation (σ) is unknown,**
 2. **Sample size (n) < 30** (small sample),
 3. Data approximately follows a **normal distribution**.
-

► Formula:

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

where s = sample standard deviation.

► **Example:**

A dietician claims that a new diet plan reduces average weight to **below 70 kg**.

A sample of **16 people** has an average weight of **68 kg** with a **sample SD = 3 kg**.

Test at **$\alpha = 0.05$** .

$$H_0 : \mu = 70, \quad H_1 : \mu < 70$$

$$t = \frac{68 - 70}{3/\sqrt{16}} = \frac{-2}{0.75} = -2.67$$

Degrees of freedom $df = 16 - 1 = 15$

From t-table, $t_{0.05,15} = -1.753$

Since **$-2.67 < -1.753$** ,

 **Reject H_0** → The diet plan significantly reduces weight.

◆ What Are “One-tailed” and “Two-tailed” Tests?

They refer to **where** (in the probability distribution) you’re testing for evidence **against the null hypothesis (H_0)** —

that is, whether you’re looking for a difference in **one direction** or **both directions**.

◆ One-tailed Test (Directional Test)

You test for a change in **only one direction** — either **greater than** or **less than** the hypothesized value.

Types:

- **Right-tailed test:** checking if the mean is **greater than** μ_0

$$H_1 : \mu > \mu_0$$

- **Left-tailed test:** checking if the mean is **less than** μ_0

$$H_1 : \mu < \mu_0$$

Example:

A factory claims the average lifetime of bulbs is **1000 hours**.

You suspect it's **less** than that.

$$H_0 : \mu = 1000, \quad H_1 : \mu < 1000$$

This is a **left-tailed test** (only the lower side of the curve matters).

**Critical z ($\alpha = 0.05$, left tail) =
-1.645**

◆ Two-tailed Test (Non-directional Test)

You test for a change in **either direction** — the mean could be **greater or less** than μ_0 .

$$H_0 : \mu = \mu_0, \quad H_1 : \mu \neq \mu_0$$

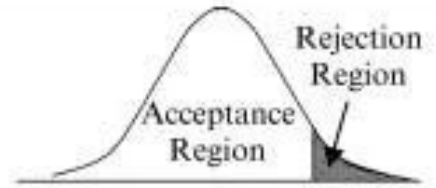
Example:

You want to check whether a machine's output **differs** from 500 ml (could be more or less).

$$H_0 : \mu = 500, \quad H_1 : \mu \neq 500$$

Critical z ($\alpha = 0.05$) = ± 1.96

This is a **two-tailed test** because deviation can occur on **both sides**.

One-Tailed Test (Left Tail)	Two-Tailed Test	One-Tailed Test (Right Tail)
$H_0 : \mu_X = \mu_0$ $H_1 : \mu_X < \mu_0$	$H_0 : \mu_X = \mu_0$ $H_1 : \mu_X \neq \mu_0$	$H_0 : \mu_X = \mu_0$ $H_1 : \mu_X > \mu_0$
		



Summary Table

Feature	One-tailed Test	Two-tailed Test
Direction of test	One direction (greater or less)	Both directions
Alternative Hypothesis (H_1)	$\mu > \mu_0$ or $\mu < \mu_0$	$\mu \neq \mu_0$
Rejection region	Entire α in one tail	$\alpha/2$ in each tail
Critical z ($\alpha = 0.05$)	± 1.645 (one side)	± 1.96 (both sides)
Example	"Is the new drug better ?"	"Is the new drug different (better or worse)?"

◆ Example: Testing a Claim about Mean

Scenario:

A company claims that the average lifetime of its LED bulbs is **10,000 hours**.

A consumer group believes the actual average is **less than 10,000 hours** and takes a **sample of 30 bulbs**.

Sample mean ($\bar{x}$) = 9,800 hours

Population standard deviation (σ) = 500 hours

Significance level (α) = 0.05

Step 1: Set up hypotheses

$$H_0 : \mu = 10000 \quad (\text{no difference})$$

$$H_1 : \mu < 10000 \quad (\text{bulbs last less than claimed})$$

Step 2: Compute test statistic

Using **z-test** (since σ is known):

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$
$$z = \frac{9800 - 10000}{500 / \sqrt{30}} = \frac{-200}{91.29} = -2.19$$

Step 3: Find critical value or p-value

For $\alpha = 0.05$ (left-tailed test), critical $z = -1.645$.

Since $z = -2.19 < -1.645$, we reject H_0 .

Step 4: Conclusion

There is **significant evidence** that the **average bulb life is less than 10,000 hours**.

◆ Interpretation

- **Reject H_0** → There's enough evidence to support the claim in H_1 .
- **Fail to Reject H_0** → Not enough evidence to prove H_1 ; H_0 may still be true.

◆ Example: Testing Whether a New Teaching Method Improves Scores

Scenario:

A professor claims that a **new teaching method** improves students' average test scores.

The **current average** score (under the old method) is **70**.

After introducing the new method to a class of 25 students, their **mean score** was **73**, with a **sample standard deviation** of **8**.

We'll test this claim at $\alpha = 0.05$ significance level.

Step 1: Define hypotheses

$$H_0 : \mu = 70 \quad (\text{no improvement})$$

$$H_1 : \mu > 70 \quad (\text{scores improved})$$

This is a **right-tailed test** (since we are testing for improvement).

Step 2: Identify known values

Symbol	Meaning	Value
$\bar{x}$	Sample mean	73
μ_0	Population mean	70
s	Sample standard deviation	8
n	Sample size	25
α	Significance level	0.05

Step 3: Compute the test statistic (t-test)

Since the **population standard deviation (σ) is unknown**, we use **t-test**:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Substitute:

$$t = \frac{73 - 70}{8/\sqrt{25}} = \frac{3}{8/5} = \frac{3}{1.6} = 1.875$$

Step 4: Determine critical value

Degrees of freedom $df = n - 1 = 24$

For a **right-tailed test** with $\alpha = 0.05$ and $df = 24$,
from t-table $\rightarrow t_{\text{critical}} = 1.711$

Step 5: Decision

Compare calculated t with critical t:

$$t_{\text{calculated}} = 1.875 > 1.711 = t_{\text{critical}}$$

 **Reject H_0**

Step 6: Conclusion

There is **significant evidence at the 5% level** that the **new teaching method improves average test scores**.

Types of Errors in Hypothesis Testing

Error Type	Description	Example
Type I Error (α)	Rejecting H_0 when it is true	False positive
Type II Error (β)	Failing to reject H_0 when it is false	False negative

Example:

- Suppose a medical test shows a person has a disease when they actually don't — that's a **Type I error**.
- A medical test fails to detect a disease that the person actually has — that's a **Type II error**.

DECISION

TRUTH

Accept H_0

Reject H_0

H_0 is true

Correct Decision P
1 - alpha

Type I Error P
alpha (Significance)

H_0 is false

Type II Error P
beta

Correct Decision P
1 - beta (Power)

H_0 = null hypothesis

P = probability

Understanding Bias and Variance

These two concepts are central to **model evaluation** and **prediction accuracy** in statistics and machine learning.

Bias: Error due to oversimplifying a model.

i.e. The model's predictions are systematically different from true values.

- **Example:**

Using a straight line to fit a highly curved dataset.

- **High bias → Underfitting**

$$\text{Bias} = E[\hat{\theta}] - \theta$$

Variance

Error due to model's sensitivity to small fluctuations in training data.

i.e. The model fits training data too closely and performs poorly on new data.

• **High variance** → **Overfitting**

Bias–Variance Tradeoff

Bias	Variance	Model Type	Problem
High	Low	Simple model	Underfitting
Low	High	Complex model	Overfitting
Balanced	Balanced	Optimal model	Good generalization

Goal: Find the balance where **total error** = $\text{bias}^2 + \text{variance} + \text{noise}$ is minimized.

Significance Levels and p -values

Significance Level (α)

- The **threshold probability** for rejecting the null hypothesis.
- Common values:
 - $0.05 \rightarrow 5\%$ chance of wrongly rejecting H_0
 - $0.01 \rightarrow 1\%$ chance of wrongly rejecting H_0

Interpretation:

If $\alpha = 0.05$, it means we are willing to accept a **5% risk** of concluding there is an effect when there is none.

p -value

- The **probability** of obtaining a test statistic at least as extreme as the one observed, assuming H_0 is true.
- Used to decide whether to reject H_0 .

p-value	Decision	Interpretation
$p \leq \alpha$	Reject H_0	Statistically significant
$p > \alpha$	Fail to reject H_0	Not statistically significant

Example:

- $\alpha = 0.05$, $p = 0.03 \rightarrow$ Reject H_0
- $\alpha = 0.05$, $p = 0.12 \rightarrow$ Fail to reject H_0

If $p < \alpha$, observed result lies in the **critical region**, meaning it's unlikely under H_0 .

Significance Level		Specification
$p > 0.05$		not significant
$p \leq 0.05$	(5%)	significant
$p \leq 0.01$	(1%)	very significant
$p \leq 0.001$	(0.1%)	highly significant

Confidence Intervals (CI)

A **confidence interval** gives a range of values within which the true population parameter is expected to lie, with a certain level of confidence.

- It provides both an **estimate** and an **uncertainty range**.

Formula:

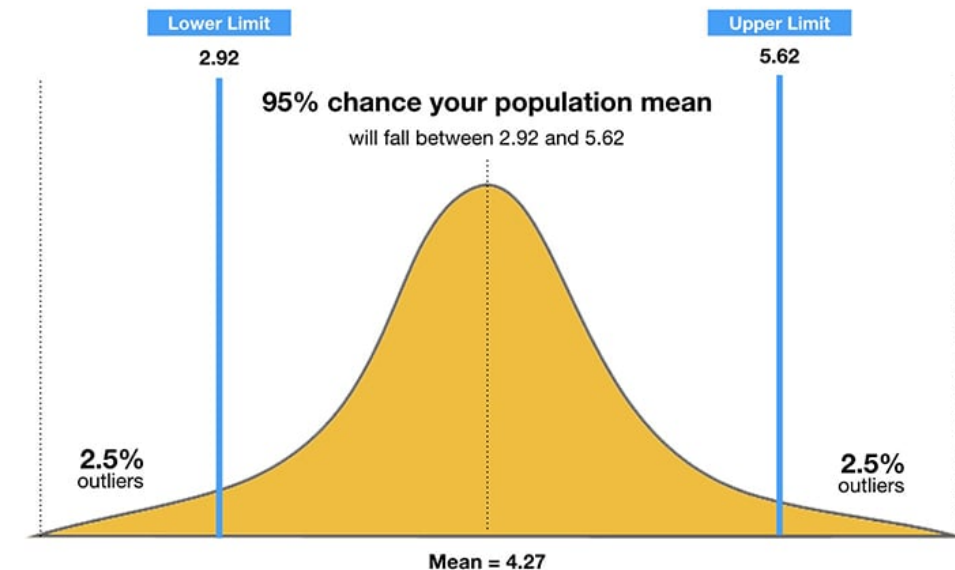
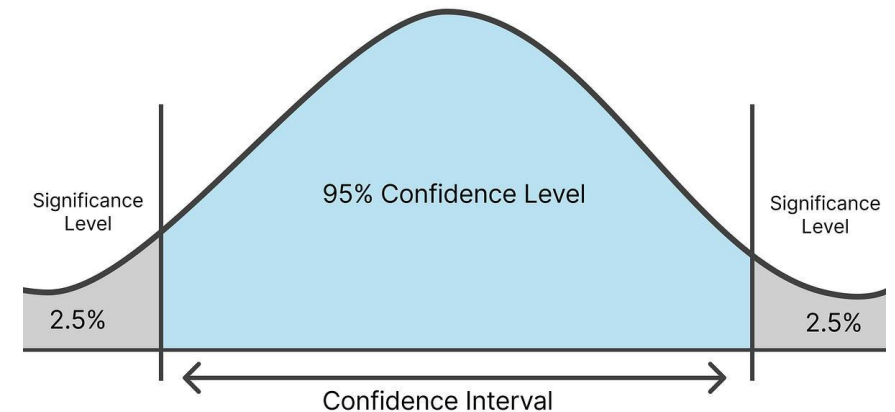
For a population mean (when σ is known):

$$CI = \bar{X} \pm Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$$

When σ is unknown (use t-distribution):

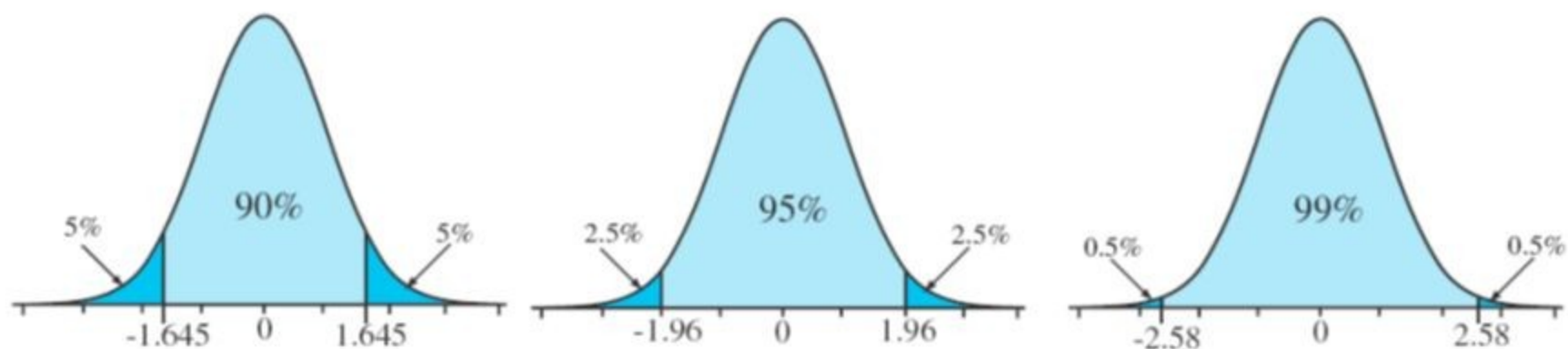
$$CI = \bar{X} \pm t_{\alpha/2, n-1} \times \frac{s}{\sqrt{n}}$$

- Informally , a confidence level does *not* represent a “ probability of being correct”; instead, it represents the frequency that the obtained answer will be accurate.



Common Levels of Confidence

Confidence level	Alpha level	Z value
$1 - \alpha$	α	$Z_{1-(\alpha/2)}$
.90	.10	1.645
.95	.05	1.960
.99	.01	2.576



Example:

A random sample of 100 students has a mean score of 70 with a standard deviation of 10.

95% confidence interval:

$$CI = 70 \pm 1.96 \times \frac{10}{\sqrt{100}} = 70 \pm 1.96$$

$$CI = (68.04, 71.96)$$

Interpretation:

We are 95% confident that the **true mean score** of the population lies between **68.04 and 71.96**.

Confidence Level vs. Significance Level

Concept	Symbol	Meaning
Confidence Level	$(1 - \alpha)$	Probability that the CI contains true parameter
Significance Level	α	Probability of rejecting a true H_0

If $\alpha = 0.05 \rightarrow$ Confidence level = 95%

Summary Table

Concept	Definition	Key Formula / Idea	Example Insight
Hypothesis Testing	Decision-making process based on sample data	t-test, z-test	Test if two groups differ significantly
Bias	Error from oversimplification	$E[\theta] - \theta$	Linear model on nonlinear data
Variance	Error from sensitivity to data	$\text{Var}(\theta)$	Overfitted model
Significance Level (α)	Threshold for rejecting H_0	Usually 0.05	5% risk tolerance
p-value	Probability of observing data under H_0	From test statistic	$p < 0.05 \Rightarrow$ significant
Confidence Interval	Range where population parameter lies	$(\bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}})$	Mean score in 95% CI (68.04, 71.96)